



School of Computer Science and Engineering [SCOPE]

CAT – I

CSE2004 – Database Management System

Course Name: B. Tech CSE

Slot: C2+TC2

Max. Marks: 30

SET-1

Answer ALL Questions (3* 10 = 30)

Q. No	Questions	Max Marks	CO	BL
1	Explain the following architectures with example: ✓ Three-Tier Client/Server architecture ✓ Two-Tier Client/Server architecture	10	1	L2
2	Design the ER diagram based on the following scenario: The ICC T20 World Cup is going to be organized by October 2021. Assume that you are given the following requirements to create a simple database for the same: • Word Cup 2021 has as many as eight teams. • Each team has a team name, country, coaches, set of players and their relevant information. • Each player belongs to only one team. • Each player has a name, their playing records (total century, total half-century, total wicket etc.) • A match is played between any two teams and has a venue, date and winning status. Further, reduce the ER diagram into a relational schema.	10	2	L3
3	Given a relation scheme R(ABCDEF) with a set of FDs $F = \{ AB \rightarrow C, c \rightarrow A, BC \rightarrow D, ACD \rightarrow B, BE \rightarrow C, CE \rightarrow FA \}$. (i) Find the minimal cover of the following set of functional dependencies. (ii) Find the closure of AB and BC.	(6+4)	2	L3

SET-2

Answer ALL Questions (3* 10 = 30)

Q. No	Questions	Max Marks	CO	BL
1	Assume that you are going to design the database of the green Vellore project. List at least five major steps that are required to set up the database.	10	1	L2
2	Design an ER diagram to track the policyholder data for Life Insurance Corporation India. LIC is offering three different types of policies Health, General and Motor insurance. It is required to maintain all customer information with their insurance. One customer can have multiple insurance policies. After maturity of the policy term policyholder will get the maturity amount. State the assumptions that you have made while designing the database management system for this application.	10	2	L2
3	Let the relation R be R (A, B, C, D, E) and the given set of FDs be F: { $AD \rightarrow C$, $ABC \rightarrow AD$, $C \rightarrow BE$, $CE \rightarrow A$, $D \rightarrow E$ }. (i) Find the minimal cover of the following set of functional dependencies. (ii) Find the closure of AD and D	(6+4)	2	L3

SET-3

Answer ALL Questions (3* 10 = 30)

Q. No	Questions	Max Marks	CO	BL
1	What is the use of data abstraction in DBMS. Explain with a scenario?	10	1	L2
2	Design an ER diagram for Codetantra in the view of the online examination system. State the assumptions that you have made while designing the database management system for this application.	10	2	L2
3	Consider the following relation R(A, B, C, D, E, F, G) and set of functional dependencies. $F = \{BCD \rightarrow A, BC \rightarrow E, A \rightarrow F, F \rightarrow G, A \rightarrow G, C \rightarrow D\}$. (i) Find the minimal cover of the following set of functional dependencies. (ii) Find the closure of BC and A.	(6+4)	2	L3